

Step-by-Step KiCad PCB Design Guide

Variable Switching DC Power Supply

STM32-Based Synchronous Buck | 1.25–30 V | 0–5 A | >88% Efficiency

This guide walks through the complete KiCad workflow for creating the PCB for the Variable Switching DC Power Supply project. Each step includes the exact KiCad action and the engineering reason behind it. First of all create a new project in KiCad named Variable Voltage Supply. Then follow the steps explained below.

Phase 1: Schematic Capture & Preparation

Goal: A clean, error-free netlist is the foundation of a good PCB.

Step 1: Assign Footprints to Every Symbol

KiCad Action:

- Open Symbol Editor or from the schematic, select a symbol → Properties (E) → Footprint → click the book icon → choose the physical package.
For this project specifically:
 - STM32F103C8T6 → Package_QFP: LQFP-48_7x7mm_P0.5mm
 - IR2104 → Package_SOIC: SOIC-8_3.9x4.9mm_P1.27mm
 - IRF540N → Package_TO_SOT_THT: TO-220-3_Vertical (or D2PAK if using SMT)
 - INA226 → Package_SOIC: SOIC-8_3.9x4.9mm_P1.27mm
 - Electrolytic caps → Capacitor_THT: CP_Radial_D10.0mm_P5.00mm (or SMD equivalents)
 - Resistors → Resistor_SMD: R_0805_2012Metric (0805 is a good balance of solderability and density)

Why this matters: The footprint links the abstract symbol to the physical copper, solder mask, and silkscreen. A wrong footprint (e.g., SOIC-8 vs. SOIC-8-N) means the part won't fit or will short. For power components like the IRF540N, the footprint determines heatsink mounting and copper area for heat dissipation.

Step 2: Add Power Flags & Labels

KiCad Action:

- Place Power Symbols (#PWR) for every voltage domain: +45V (DC bus), +3V3, +5V, GND, VIN_AC_L, VIN_AC_N.
Use Global Labels (Ctrl+L) for nets that cross sheets: PWM_H0, PWM_LO, I2C_SCL, I2C_SDA, V_SENSE, I_SENSE_ALERT.
Place Net Class directives on critical nets if needed.

Why this matters: KiCad's ERC (Electrical Rule Check) needs power flags to know that nets are driven, not floating. Without them, you'll get hundreds of false errors. Global labels keep the schematic readable while ensuring KiCad knows these are the same net across pages. For this project, SW_NODE must be a single, unambiguous net — any label error here means the gate driver won't connect to the MOSFET.

Step 3: Run ERC (Electrical Rule Check)

KiCad Action:

- Inspect → Electrical Rules Checker → Run ERC
Fix all errors (red) and warnings (yellow).
Common issues to fix:
 - Unconnected pins: Tie unused GPIOs to GND or mark as 'Not Connected' (X tool).
 - Power pins not driven: Add power flags.
 - Conflicting drivers: Two outputs tied together.

Why this matters: ERC catches schematic errors that become PCB shorts or opens. A missed unconnected pin on the IR2104's /SD pin means your shutdown protection won't work — a catastrophic safety failure. A short between +45V and SW_NODE in the schematic will become a dead MOSFET in real life.

Step 4: Annotate & Generate Netlist

KiCad Action:

- Tools → Annotate Schematic → Annotate Automatically (or manually if you prefer).
Tools → Update PCB from Schematic (or File → Export → Netlist if using older workflow).

Why this matters: Annotation gives every part a unique reference designator (R1, R2, C1, U1). The netlist is the 'DNA' of the board — it tells KiCad which pins must be connected. Without it, the PCB editor has no idea that pin 1 of R1 connects to pin 3 of U2.

Phase 2: Board Setup & Stackup

Goal: Define the physical reality of your board before placing a single component.

Step 5: Set Board Edge & Origin

KiCad Action:

- Switch to PCB Editor.
File → Board Setup → Board Editor → Origin → set to lower-left corner.
Use the Edge.Cuts layer → draw a closed rectangle with the line tool.
Suggested size: 100 mm × 80 mm (adjust based on transformer and heatsink).

Why this matters: The edge.cuts layer defines the physical boundary. The origin is your coordinate reference for manufacturing files. A board too small means components collide; too large wastes money and increases loop inductance. For this project, the transformer T1 and heatsink for Q1/Q2 will dominate the mechanical envelope.

Step 6: Configure the 4-Layer Stackup

KiCad Action:

- File → Board Setup → Board Stackup → Physical Stackup
Set Copper Layers = 4
Configure layers as shown in the table below.

Set dielectric thickness: Prepreg 0.1 mm, Core 1.2 mm (standard for most fabs).

Set copper weight: 1 oz outer, 0.5 oz inner (upgrade to 2 oz outer if >3 A continuous).

Layer	Name	Type	Purpose
Top	F.Cu	Signal	MCU, control ICs, analog sense, gate drive
Inner 1	In1.Cu	Ground	Solid ground plane — never broken
Inner 2	In2.Cu	Power	Power distribution (+45V, +3V3, +5V, SW node)
Bottom	B.Cu	Signal + Power	MOSFETs, high-current traces, heatsink copper

Why this matters: This stackup is the backbone of your EMC and thermal performance. L2 (solid ground) provides a low-impedance return path directly beneath every signal trace. At 200 kHz, return currents follow the path of least inductance — which is directly under the signal trace. A solid plane makes this automatic. A split plane forces return currents to detour, creating large loop antennas that radiate EMI. L3 distributes high-current rails without stealing space from L2. Top/Bottom split keeps noisy power stuff away from sensitive analog/MCU.

Step 7: Enter Design Rules (DRC)

KiCad Action:

- File → Board Setup → Design Rules → Constraints
Set the values shown in the table below.
File → Board Setup → Design Rules → Net Classes → Create classes as shown in the Net Classes table.

Custom Rules: Go to Rules → Custom Rules and add the AC creepage rule.

Rule	Value	KiCad Field
Minimum track width	0.15 mm	Minimum track width
Minimum clearance	0.15 mm	Minimum clearance
Minimum via drill	0.3 mm	Minimum via diameter (drill)
Minimum via size	0.6 mm	Minimum via diameter (total)

Net Classes:

Net Class	Track Width	Clearance	Via Size	Why
Default	0.25 mm	0.15 mm	0.6 mm	Signals, I2C, gate drive
Power_High	2.0 mm	0.3 mm	1.0 mm	+45V bus, SW node, Vout ($\geq 2A$)
Power_Low	0.5 mm	0.15 mm	0.6 mm	+3V3, +5V logic rails
AC_Mains	1.5 mm	2.0 mm	1.2 mm	AC input before transformer

Custom Rule (AC Creepage):

```
(rule AC_creepage
  (condition "A.NetClass == 'AC_Mains' &&
B.NetClass == 'Default'")
  (constraint clearance (min 4.0mm))
)
```

Why this matters: These rules are your guardrails. KiCad will flag violations in real-time. The 2 mm per amp rule means your Power_High class at 2 mm handles ~1 A comfortably; for the 5 A output path, you'll

need 10 mm copper pours (not traces). The 4 mm AC-to-DC creepage is a safety regulation (IEC 60950 / UL) — at 230 V AC, insufficient creepage can cause tracking (carbonization) across the PCB surface over time, leading to fire or shock hazard.

Phase 3: Component Placement

Goal: Placement determines 80% of your board's performance. Routing only connects what placement dictates.

Step 8: Place the Transformer & AC Input Section

KiCad Action:

- Place T1 (toroidal transformer) near the board edge where AC enters.
Place F1, MOV, CMC, Cx in a line leading to T1 primary.
Keep all AC primary components in one corner, separated from the DC side by a 4 mm gap (creepage).

Why this matters: This is your EMI filter at board entry point (per your rules). The CMC and X-cap must be right at the entry to catch conducted noise before it spreads. The 4 mm separation between AC and DC domains prevents arcing and satisfies safety standards. The transformer provides isolation — respect it by keeping primary and secondary copper regions physically apart.

Step 9: Place the Power Stage (Q1, Q2, L1, C2)

KiCad Action:

- Place Q1 (high-side) and Q2 (low-side) adjacent to each other.
Place L1 so its terminal is within 10 mm of the Q1/Q2 common node (SW node).
Place C2 directly adjacent to the SW node and Q2's drain/source.
Orient Q1 and Q2 so their tab/drain pins face each other — this minimizes the copper distance between them.

Why this matters: This is your switching loop (Q1 → L1 → C2 → GND). Your rule says loop area < 1 cm². Loop area = inductance = radiated EMI + voltage spikes. A large loop means high di/dt (200 kHz switching) induces voltage spikes that can avalanche your MOSFETs. By placing these four components in a tight cluster, you make the loop physically small. C2 must be adjacent because it supplies the AC

component of the inductor current — if it's far away, the path through the capacitor has high inductance and the ripple voltage increases.

Step 10: Place the Gate Driver (IR2104)

KiCad Action:

- Place IR2104 within 20 mm of Q1 and Q2 gate pins.
Place the bootstrap diode and capacitor (D_boot, C_boot) right next to the IR2104.

Why this matters: Gate drive traces carry high peak currents (amps) to charge/discharge MOSFET gate capacitance in nanoseconds. Long traces have inductance, which causes ringing and slow switching — increasing switching losses. The bootstrap capacitor must be close because it recharges every cycle; long traces add resistance and inductance that can cause the high-side supply to droop below UVLO during switching.

Step 11: Place the MCU & Control Section

KiCad Action:

- Place STM32F103C8T6 on the opposite side of the board from the power stage.
Place INA226, OLED connector, encoder near the STM32.
Place all 3.3 V decoupling capacitors (100 nF) within 3 mm of every VDD pin.

Why this matters: This is partition by placement, not by plane cuts. The STM32's ADC reads millivolt-level signals (voltage divider, current sense). The SW node switches 45 V at 200 kHz with nanosecond edges. If the MCU sits next to the SW node, capacitive coupling and magnetic induction will corrupt your ADC readings. The 3 mm decoupling rule ensures the capacitors have minimal trace inductance to the pin — at 200 kHz, even 5 mm of trace adds enough inductance to make the capacitor ineffective.

Step 12: Place the Output Terminals & Sense Points

KiCad Action:

- Place output banana jacks / screw terminals at the board edge.
Place the voltage divider resistors close to the output terminals, but route a Kelvin sense trace back to the STM32.
Place the INA226 near the current shunt resistor.

Why this matters: The voltage divider is your feedback path. If it's far from the output, trace resistance adds error. The Kelvin sense (separate traces for force and sense) eliminates voltage drop in the high-current output traces from affecting your regulation. The INA226 must be near the shunt to minimize noise pickup on the differential sense lines.

Phase 4: Routing Strategy

Goal: Route critical nets first, then fill planes, then clean up.

Step 13: Route the Switching Loop (Top Priority)

KiCad Action:

- Route Q1 drain → +45V pour (short, wide).
Route Q1 source / Q2 drain → SW node (this is the critical node — keep it as a copper pour, not a trace).
Route Q2 source → GND via multiple vias to L2.
Route SW node → L1 (short, wide trace or pour).
Route L1 → C2+ → Output (short, wide).
Route C2- → GND via multiple vias to L2.

Why this matters: This loop carries pulsed current at 200 kHz. Every millimeter of trace adds inductance. Using copper pours instead of traces reduces resistance and inductance. Multiple vias to L2 (ground) reduce via inductance — one via has ~1 nH; four in parallel have ~0.25 nH. This directly impacts your output ripple and MOSFET switching stress.

Step 14: Route Gate Drive Signals

KiCad Action:

- Route IR2104 HO → Q1 gate (short, ~0.3 mm trace).
Route IR2104 LO → Q2 gate (short, ~0.3 mm trace).
Add a series resistor (10–22 Ω) close to each gate pin for damping.
Route IR2104 GND → Power GND via a single, short trace to a quiet ground point (avoid the switching loop return path).

Why this matters: Gate traces are antennas. Long gate traces pick up noise from the SW node and can falsely turn on MOSFETs. The series resistor dampens ringing caused by trace inductance resonating with gate capacitance. The IR2104's GND must return to a clean ground — if you tie it into the high-current switching return, voltage drops across that ground will shift the IR2104's reference and cause erratic switching.

Step 15: Create the Solid Ground Plane (L2)

KiCad Action:

- Select In1.Cu layer.
Use the Zone tool (Ctrl+Shift+Z) → select GND net → draw a polygon covering the entire board (inside edge cuts).
Set Pad connection: Thermal relief for soldering, Via connection: Direct.
Do NOT draw any lines or keepouts that split this plane.

Why this matters: This is your solid, unbroken ground plane. It provides: (1) Low impedance return path for every signal (minimizes loop area), (2) Shielding between the noisy power stage (Top) and the power distribution layer (L3), (3) Heat spreading for the bottom-side MOSFETs. A split plane (e.g., 'analog GND' vs. 'digital GND') forces return currents to find long detours, creating slot antennas that radiate at 200 kHz and harmonics. Your STM32's ADC will see this as noise.

Step 16: Create the Power Plane (L3)

KiCad Action:

- Select In2.Cu layer.
Draw zones for: +45V (large pour under the power stage), +3V3 (pour under the MCU region), +5V (small pour near the regulator).
Keep SW node as a top-layer pour only — do not put it on L3 (it would couple noise into everything).

Why this matters: L3 distributes power without consuming top-layer routing space. The +45V pour acts as a low-inductance reservoir for the switching current. Keeping SW node off L3 prevents 45 V square waves from capacitively coupling into the power plane and radiating. The +3V3 pour under the MCU provides a clean, low-impedance supply.

Step 17: Route Sensitive Analog & I2C

KiCad Action:

- Route PA0 (V_SENSE) and PA1 (TEMP_SENSE) as short, thin traces (0.25 mm) away from the SW node.
Route I2C (PB6/PB7) as a differential pair (0.25 mm width, 0.25 mm spacing) with pull-ups near the STM32.
Add a ground guard ring around the voltage divider and sense traces (connect to L2 with vias).

Why this matters: The voltage divider output is a high-impedance node. A 0.5 V induced noise spike (easy to pick up near a 45 V SW node) becomes a 4% error in your 12 V reading. Guard rings intercept capacitive coupling and shunt it to ground. I2C is slow (400 kHz) but sensitive to crosstalk — keeping SDA and SCL as a pair with consistent geometry prevents skew and noise.

Step 18: Add Ground Stitching Vias

KiCad Action:

- Place vias around the board perimeter, spaced every 10–15 mm.
Add a via grid under the Q1/Q2/L1/C2 cluster (spacing ~5 mm).
Add vias near every decoupling capacitor pad.

Why this matters: Stitching vias tie the top and bottom ground pours to the solid L2 plane, creating a Faraday cage that contains EMI. Without them, the top and bottom layers act as patch antennas at 200 kHz. The dense via grid under the power stage provides a low-impedance path for the high-frequency return current, reducing ground bounce.

Step 19: Add the Ferrite Bead on Debug Ground

KiCad Action:

- Place a ferrite bead in series with the GND pin of your SWD/debug connector.
Route the debug connector's GND through the bead to the main ground plane.

Why this matters: When you connect a debugger or USB cable, you create a ground loop between your PC earth and the board ground. The

ferrite bead blocks high-frequency noise from traveling out on the cable (preventing your PC from becoming an antenna) while allowing DC/low-frequency return current for the debugger to function. This is a common source of EMI failures in lab prototypes.

Phase 5: Verification & Output

Goal: Verify correctness and generate manufacturing files.

Step 20: Run DRC (Design Rule Check)

KiCad Action:

- Inspect → Design Rules Checker → Run DRC
Fix all errors (unconnected nets, clearance violations, track width violations).
Pay special attention to:
 - AC-to-DC clearance: Ensure no copper crosses the 4 mm boundary.
 - Switching loop: Verify Q1→L1→C2→GND path is short and wide.
 - Decoupling proximity: Check that 100 nF caps are within 3 mm of STM32 VDD pins.

Why this matters: DRC is your last line of defense before fabrication. An unconnected net means a board that doesn't work. A clearance violation on AC mains means a safety hazard. A narrow trace on the 5 A output means a fire risk (trace fusing). KiCad's DRC enforces the rules you set in Step 7 — but it only knows about what you've drawn.

Step 21: Run the 3D Viewer & Mechanical Check

KiCad Action:

- View → 3D Viewer
Verify:
 - Transformer height clears enclosure lid.
 - Heatsink on Q1/Q2 doesn't short to nearby components.
 - Encoder and OLED connector are accessible from the front panel.
 - AC input terminals are recessed and finger-safe.

Why this matters: A board that works electrically but doesn't fit the enclosure is useless. The 3D viewer catches height conflicts (e.g., electrolytic cap hitting the lid) and placement errors (e.g., connector blocked by the heatsink). For this project, the TO-220 MOSFETs with heatsinks are the tallest components.

Step 22: Generate Manufacturing Outputs

KiCad Action:

- File → Fabrication Outputs → Gerbers
Layers: F.Cu, In1.Cu, In2.Cu, B.Cu, F.SilkS, F.Mask, B.Mask, Edge.Cuts
File → Fabrication Outputs → Drill Files (.drl)
File → Fabrication Outputs → IPC-D-356 (netlist for electrical test at fab)
File → Fabrication Outputs → Bill of Materials (BOM)

Why this matters: Gerbers are the universal language of PCB fabrication. The drill file tells the CNC where to make holes. The IPC netlist allows the fab to electrically test every board for shorts/opens before shipping — catching manufacturing defects. The BOM ensures you order the right parts.

Summary Checklist

Phase	Key Deliverable	Your Design Rule Check
Schematic	ERC clean, footprints assigned	All symbols linked to correct packages
Stackup	4-layer, 1.6 mm total	L2 = solid ground, L3 = power
Placement	Transformer at entry, MCU far from power	Partition by placement, 4 mm AC-DC gap
Power Stage	Q1/Q2/L1/C2 cluster	Loop area < 1 cm ²
Routing	Short gate drives, Kelvin sense	2 mm per amp, 0.15 mm signal min
Planes	Solid L2 ground, no splits	Return currents flow beneath signals
Stitching	Vias every 10–15 mm perimeter	Faraday cage, low ground impedance
DRC	Zero errors, zero warnings	Clearance, creepage, width all pass
Outputs	Gerbers, drill, BOM, IPC netlist	Ready for fabrication

Next step: Start with Step 1 (footprints) and work through to Step 5 (netlist). Once in the PCB editor, follow the placement order strictly: AC input → Power stage → Gate driver → MCU → Output. This sequence prevents you from 'painting yourself into a corner' where the critical switching loop becomes too large.

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